UNIVERSITY OF MORATUWA
Faculty of Engineering
Department of Electronic and Telecommunication Engineering
B.Sc. Engineering
Semester 6 Examination

EN3330 INTRODUCTION TO ENGINEERING OPTIMIZATION

Time allowed: **Two (2)** hours June 2024

INSTRUCTIONS TO CANDIDATES:

- This paper contains 4 questions on 5 pages.
- This examination accounts for **30**% of the module assessment. The total maximum mark attainable is **100**. The marks assigned for each question & sections there of are indicated in brackets.
- This is a closed-book examination.
- Electronic/communication devices are not permitted. Only equipment allowed is a calculator approved and labeled by the Faculty of Engineering.
- Derivations are not required if they are not explicitly requested for in the question.
- Assume reasonable values for any data not given in or with the examination paper. Clearly state such assumptions.
- If you have any doubt as to the interpretation of the wording of a question, make your own decision, and clearly state it.
- All questions carry equal marks.
- Answer **ALL** questions.
- Keep your answers **brief** and to the point.

ADDITIONAL MATERIAL:

- No additional materials are provided.

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Question 1.

(a) Five critical points of a function $f(x)$, $\{x \in \mathbb{R} : -1.5 \leq x \leq 13\}$ is shown in Figure Q1.1. State the given critical points A, B, C, D and E. [5 marks]

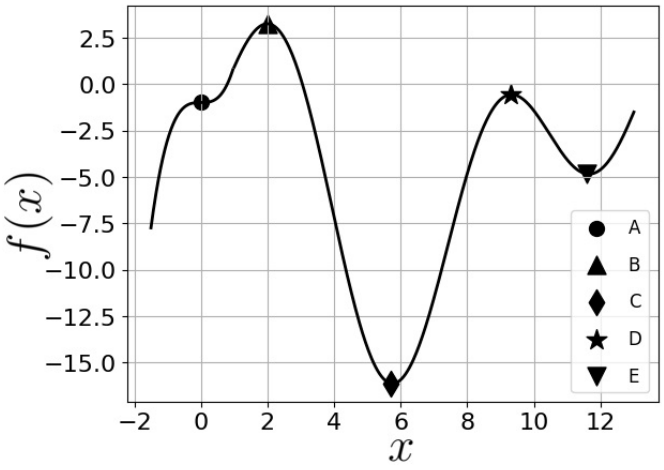


Figure Q1.1: Critical points of a function $f(x)$.

(b) Consider the following optimization problem:

$$\begin{aligned} \underset{\boldsymbol{x}}{\text{minimize}} \quad & f(\boldsymbol{x}) = \left(x_1 - \frac{4}{3}\right)^n + (x_2 - 5)^n, \\ \text{such that} \quad & x_1 + 2x_2 - 8 \geq 0, \\ & 2x_1 + x_2 - 6 \leq 0, \\ & -x_1 \leq 0, \\ & -x_2 \leq 0. \end{aligned}$$

i. Draw a sketch of the feasible region. [4 marks]

ii. For $n = 2$, sketch both the minimum of $f(\boldsymbol{x})$ and the feasible region in a same diagram. [6 marks]

iii. For the question Q1.ii, list active and inactive constrains. [2 marks]

(c) State first-order necessary condition (FONC) and second-order necessary condition (SONC) for a local minimum of a multi-variable function. [2 marks]

(d) Consider the function $f(\boldsymbol{x}) = \frac{x_1^3}{3} - \frac{9x_1^2}{2} + 20x_1 + x_2^2 - 6x_2 + 9$. Identify critical points and determine they are minima or maxima or neither minimum nor maximum. [6 marks]

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Question 2.

(a) Consider the following Euclidean norm minimization problem:

$$\begin{aligned} &\text{minimize} \quad ||\boldsymbol{x}-\boldsymbol{d}||_2^2 \\ &\text{subject to} \quad \boldsymbol{a}^T\boldsymbol{x} = b \end{aligned}$$

where $\boldsymbol{d}, \boldsymbol{a} \in \mathbb{R}^n$ and $b \in \mathbb{R}$ are known parameters, whereas $\boldsymbol{x} \in \mathbb{R}^n$ denotes the optimization variable.

i. Show that the above problem is equivalent to

$$\begin{aligned} &\text{minimize} \quad ||\boldsymbol{y}||_2^2 \\ &\text{subject to} \quad \boldsymbol{a}^T\boldsymbol{y} = c \end{aligned}$$

where $c = b - \boldsymbol{a}^T\boldsymbol{d}$. [2 marks]

ii. Now use purely geometric arguments to find the optimal $\boldsymbol{x} = \boldsymbol{x}^*$ which solves the original problem. [5 marks]

(b) A certain energy minimization problem assumes the following unconstrained form

$$\text{minimize} \quad \sum_{k=1}^{m} \beta_k ||\boldsymbol{z}-\boldsymbol{b}_k||_2^2$$

where $\boldsymbol{z} \in \mathbb{R}^n$ is the optimization variable and $\beta_k > 0,\ \boldsymbol{b}_k \in \mathbb{R}^n,\ k = 1, 2, \ldots, m$, are known parameters.

i. Capitalizing on the observation

$$||\boldsymbol{z}-\boldsymbol{b}_k||_2^2 = \boldsymbol{z}^T\boldsymbol{z} - 2\boldsymbol{z}^T\boldsymbol{b}_k + \boldsymbol{b}_k^T\boldsymbol{b}_k, \quad k = 1, 2, \ldots, m,$$

show that the optimum solution is given by [12 marks]

$$\boldsymbol{z}^* = \frac{\sum_{k=1}^{m} \beta_k \boldsymbol{b}_k}{\sum_{k=1}^{m} \beta_k}.$$

(c) Moreover, assuming that $\boldsymbol{b}_k,\ k = 1, 2, \ldots, m \leq n$, are orthonormal vectors[1], solve the following optimization problem

$$\begin{aligned} &\text{maximize} \quad \sum_{k=1}^{m} \beta_k ||\boldsymbol{z}^*-\boldsymbol{b}_k||_2^2 \\ &\text{subject to} \quad \sum_{k=1}^{m} \beta_k = 1 \end{aligned}$$

where $\beta_k > 0,\ k = 1, 2, \ldots, m$, are the optimization variables. [6 marks]

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[1]To be precise, this implies $\boldsymbol{b}_k^T\boldsymbol{b}_\ell = \begin{cases} 1 & \text{if } k = \ell \\ 0 & \text{if } k \neq \ell \end{cases}$.

Question 3.

(a) A PCB manufacturing company manufactures three types of PCBs namely type A, type B and type C. Suppose that number of PCBs that manufacture for types A, B and C are x_A, x_B and x_C, respectively. The PCB manufacturing company wants to maximize the profit which is given by

$$P(x_A, x_B, x_C) = (60\ x_A + 20\ x_B(x_B - 10) + 10\ x_C) \times 10^3.$$

For one labor working hour, it can be manufacture either one type A PCB or one type B PCB. Manufacturing a type C PCB requires two labor working hours. The company allocated 120 labor working hours for PCB manufacturing and it is required to utilize all available labor working hours.
To manufacture a type B PCB, it requires 5 minutes of machine hours. However, for a type A PCB, it requires twice the machine hours needed for a type B PCB, and for a type C PCB, it requires three times the machine hours needed for a type B PCB. The total available machine hours are 950 and it is required to utilize all available machine hours.
The cost that required to manufacture a type B PCB is half the cost of a type A PCB. Further, to manufacture a type C PCB, it costs one third of the cost of a type B PCB. It is given that the cost to manufacture a type A PCB is $30,000$ Rupees. The maximum budged available is 2.5×10^6 Rupees.

i. Formulate an optimization problem to find the number of PCBs of types A, B and C that need to manufacture to maximize the profit of the company. Note that it is required to manufacture all types of PCBs. [5 marks]

ii. Write down the Lagrangian function and KKT conditions (primal feasibility, dual feasibility and complementary slackness). [5 marks]

iii. Use the KKT conditions to solve the optimization problem. Find the feasible values of x_A, x_B and x_C, which maximize profit (In your calculation for the sake of simplicity, ignore the conditions that values of x_A, x_B, and x_C are non-negative). [10 marks]

iv. If the company decided to manufacture only a single type of PCB (either type A or type B or type C). Which type of PCB would maximize the profit? [5 marks]

Continued ...

Question 4.

(a) Consider the following unconstrained optimization: problem

$$\underset{\boldsymbol{x}}{\text{minimize}} \quad f(\boldsymbol{x}) = \frac{1}{2}\,\boldsymbol{a}^T\boldsymbol{x},$$

where $\boldsymbol{x} = [x_1^2\ x_2^2\ x_3^2]^T$ and $\boldsymbol{a} = [a_1\ a_2\ a_3]^T$.

i. Suppose that Gradient descent algorithm is used to obtain the solution of the above optimization problem. Let $x_i^{(k)}$ is the updated value of variable x_k at $k-$th iteration. Write down expressions for $x_1^{(k+1)}$, $x_2^{(k+1)}$ and $x_3^{(k+1)}$ for step size α. [5 marks]

ii. Compute rate of change $\dfrac{x_i^{(k+1)} - x_i^{(k)}}{x_i^{(k)}}$ for all three variables. [3 marks]

iii. It is given that $a_1 = 10^{-4}$, $a_2 = a_3 = 1$ and starting point as $\boldsymbol{x}^{(0)} = [5\ 5\ 5]^T$. Comment on progress of the Gradient descent algorithm in reaching to the global minimum. [5 marks]

iv. Propose two first order methods which perform better than the Gradient descent for the scenario given in Q4.iii. [2 marks]

(b) Table Q4.1 shows five data samples which are generated by multivariate normal distribution. Suppose cross-entropy method is used to obtain the maximum of the function $f(\boldsymbol{x}) = 5x_1^2 - x_2^2$.

Table Q4.1: Data samples.

Index (i)	1	2	3	4	5
$x_{1,i}$	4	8	6	10	12
$x_{2,i}$	5	25	15	10	0

i. Suppose three elite samples are used to determine the maximum of the function $f(\boldsymbol{x})$. Which three samples are used ? What is the reason behind your selection ? [2 marks]

ii. Compute the mean vector and covariance matrix for the next iteration. [8 marks]

- End of Question Paper -